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Reading and writing impairments in Spanish-speaking individuals with primary progressive aphasia: A single-case series study

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ABSTRACT

Reading and writing impairments are frequent in primary progressive aphasia (PPA), yet their characterization in Spanish-speaking populations remains limited due to language-specific orthographic properties. The present study examined reading and writing impairments in 17 Spanish-speaking participants diagnosed with PPA (9 logopenic, 4 semantic, and 4 nonfluent), with the aim of identifying patterns of acquired dyslexia and dysgraphia across PPA variants.

A single-case series design was employed, including individuals diagnosed with PPA and matched healthy control subjects. Participants were assessed using tasks of word and nonword reading, loanword reading, visual lexical decision with pseudohomophones, and writing to dictation of words and nonwords.

Results showed that 94% of participants presented acquired dyslexia and 75% acquired dysgraphia. Both impairments were observed across all PPA variants. Regarding reading profiles, five cases of surface acquired dyslexia, two of phonological dyslexia, three of mixed dyslexia, and six unspecified profiles were identified. In writing, one case of surface acquired dysgraphia, one phonological, four mixed, and six unspecified profiles were observed.

These findings provide relevant evidence regarding the presence and heterogeneity of acquired dyslexia and dysgraphia in Spanish-speaking individuals with PPA, and demonstrate that these patterns can be systematically identified when assessment procedures account for the linguistic characteristics of Spanish and the clinical features of this population.

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1. Introduction

Reading and writing are complex language-related cognitive abilities that can be selectively disrupted in neurodegenerative disorders. In primary progressive aphasia (PPA), a clinical syndrome associated with frontotemporal lobar degeneration and Alzheimer's disease and characterized by a prominent and progressive language impairment (Gorno-Tempini et al., 2011; Mesulam, 2001, 2003), difficulties in these domains have been consistently reported. In the semantic variant of PPA (svPPA), surface dyslexia and surface dysgraphia constitute supportive diagnostic features (Gorno-Tempini et al., 2011). Reading and writing impairments have also been documented in the logopenic variant (lvPPA) and the non-fluent or agrammatic variant (nfvPPA), despite not being included among their diagnostic criteria (Brambati et al., 2009; Henry et al., 2016; Neophytou et al., 2019; Petroi et al., 2014; Rohrer, Ridgway, et al., 2010; Rohrer, Ridgway, et al., 2010). However, most research on PPA has been conducted in English-speaking contexts, and studies examining these impairments in Spanish-speaking individuals remain scarce. Given that differences across orthographic systems preclude the direct generalization of findings from other languages, this gap has limited the current understanding of patterns of acquired dyslexia and acquired dysgraphia in Spanish and raises important questions regarding their assessment within current diagnostic frameworks.

The diagnostic consensus proposed by Gorno-Tempini et al. (2011) recommends the use of reading and writing to dictation tasks including regular words, irregular words, and nonwords to identify patterns of acquired surface dyslexia and dysgraphia. In Spanish, this approach poses methodological challenges, as orthographic consistency for reading is nearly perfect and most words can be accurately read through grapheme to phoneme conversion (Valle Arroyo & Vega, 1995). Truly irregular words are therefore rare. Consequently, alternative tasks have been developed in Spanish-speaking contexts to detect surface dyslexia, including visual lexical decision with pseudohomophones, homophone comprehension tasks, reading of words with omitted stress marks, and reading of loanwords, which have been widely used in individuals with aphasia due to focal lesions (Cuetos, 1999; Difalcis et al., 2021; Ferreres et al., 2005, 2012; Iribarren et al., 1996). Other patterns of reading impairment, such as phonological dyslexia, are typically identified using classical tasks, including nonword reading (Difalcis et al., 2021).

Despite these methodological advances, research specifically examining reading deficits in Spanish-speaking individuals with PPA remains limited. To date, only four studies have directly addressed reading impairments in this population, two single-case reports (Aguillón-Solís et al., 2019; Morello-García et al., 2021) and two group studies (Matías-

Guiu et al., 2017; Matías-Guiu et al., 2019). Aguillón-Solís et al. (2019) reported alterations in verb and noun reading comprehension in a participant with nfvPPA, identifying stimulus exposure time as the primary factor affecting performance; however, the findings were not interpreted within established subtypes of acquired dyslexia. In contrast, Morello-García et al. (2021) described a participant with svPPA who exhibited a profile consistent with surface dyslexia and surface dysgraphia, identified through word and nonword reading tasks, latency measures, visual lexical decision with pseudohomophones, and writing to dictation. In a group study, Matías-Guiu et al. (2017) examined 35 Spanish-speaking individuals with PPA and reported reading impairments across all variants relative to controls. Cases diagnosed with svPPA showed greater difficulties in tasks with higher lexical demands, whereas individuals with the lvPPA and nfvPPA demonstrated deficits in phonological tasks and nonword reading. A subsequent study by the same group (Matías-Guiu et al., 2019) included reading tasks and reported relevant findings; however, its primary objective was not the identification of acquired dyslexia subtypes. Taken together, these studies document reading impairment in Spanish-speaking individuals with PPA; however, their limited number, methodological heterogeneity, and variable theoretical focus do not allow for a systematic characterization of acquired dyslexia subtypes.

In addition to the challenges affecting reading, writing introduces a different type of complexity. The characterization of Spanish as a transparent language primarily applies to reading. In contrast, writing presents important asymmetries due to heterographic homophony, whereby certain phonemes may be represented by more than one grapheme. For example, the phoneme /s/ can be spelled as *s*, *c*, or *z* in Rioplatense Spanish, the variety spoken in Argentina (Ferreres et al., 2005; Signorini & Borzone de Manrique, 2003). Of the 22 phonemes in Spanish, 14 correspond uniquely to a single grapheme, whereas eight may be represented by two or three alternative graphemes. In addition, the letter *h* does not correspond to any phoneme. Consequently, Spanish cannot be considered fully transparent for writing, as the correct orthographic form of many words cannot be derived solely through phoneme to grapheme conversion rules. Moreover, Spanish includes heterographic homophones, that is, words that share the same pronunciation but differ in writing and meaning, such as *sabia* (“wise”, feminine form) and *savia* (“sap”).

Although surface dysgraphia can in principle be identified in Spanish-speaking individuals using writing to dictation tasks recommended by the diagnostic consensus (Gorno-Tempini et al., 2011), only one study to date has reported a clearly delineated dysgraphic profile in this population (Morello-García et al., 2021). A small number of additional

publications have mentioned the presence of dysgraphia in Spanish-speaking individuals with PPA; however, these studies either did not specify the pattern of impairment or did not provide sufficient detail regarding participants' performance to allow for its characterization (Donoso et al., 2008; Garzón Avellaneda et al., 2011; Reyes Nava, 2006).

The challenges posed in Spanish-speaking contexts are particularly relevant for individuals with svPPA. Although surface dyslexia and surface dysgraphia constitute supportive diagnostic features of this variant (Gorno-Tempini et al., 2011), their identification in Spanish is not straightforward. This constraint is reflected not only in the limited number of reports, but also in current assessment practices. For example, the Spanish version of the neuropsychological battery included in the FTLD module of the Uniform Data Set does not incorporate formal word reading measures, in contrast to the English version (National Alzheimer's Coordinating Center & FTLD Work Group of the ADC Program, 2015).

With respect to the lvPPA and nfvPPA, systematic research in Spanish examining reading and writing impairments remains lacking. The only study reporting detailed reading profiles that included these variants is that of Matías-Guiu et al. (2017). As for writing, no systematic publications have characterized dysgraphic patterns in these variants. However, studies conducted in other languages have consistently reported the presence of both acquired dyslexia and dysgraphia in lvPPA and nfvPPA. These impairments have been primarily associated with phonological dyslexia (Brambati et al., 2009; Neary et al., 1998; Ripamonti et al., 2017; Rohrer, Ridgway, et al., 2010; Rohrer, Ridgway, et al., 2010) and phonological dysgraphia (Henry et al., 2016; Neary et al., 1998; Neophytou et al., 2019; Shim et al., 2012). Nevertheless, individuals with these variants have also been reported to present other dyslexic and dysgraphic patterns (see, for example, Faria et al., 2013; Rohrer, Rossor, & Warren, 2010; Sepelyak et al., 2011; Tree et al., 2005; Watt et al., 1997).

In this context, the present study examines reading and writing impairments in Spanish-speaking individuals diagnosed with PPA. Specifically, it aims to identify variant specific patterns of acquired dyslexia and dysgraphia and to determine whether these profiles can be systematically identified when assessment procedures are adapted to the characteristics of Spanish orthography.

2. Methods

2.1. Design and participants

A single-case series methodology with a functional inclusion criterion was employed (Shallice, 2015), together with a case-control design (Crawford & Garthwaite, 2005). Seventeen individuals with a diagnosis of PPA were included based on this functional criterion, and each participant's performance was compared with that of healthy controls.

Participants were recruited using a non-probabilistic, purposive sampling strategy (Hernández-Sampieri et al., 2014, pp. 170–191) from the Department of Cognitive Neurology, Neuropsychology, and Neuropsychiatry at FLENI and from the

Cognitive Neurosciences Area of the Buenos Aires Institute of Neuroscience, both located in Buenos Aires, Argentina. All subjects had a diagnosis of PPA established by specialists at the participating healthcare institutions.

Inclusion criteria for the participant sample were: (1) a clinical diagnosis of PPA according to the criteria proposed by Gorno-Tempini et al. (2011); (2) being a native speaker of Rioplatense Spanish; (3) age between 50 and 90 years; and (4) literacy. Exclusion criteria included: (1) a history of acquired neurological conditions (e.g., stroke, traumatic brain injury) or neurodevelopmental disorders; and/or (2) uncorrected basic visual or auditory impairments.

The participant group comprised 17 individuals (14 women and 3 men), with a mean age of 67.8 years ($SD = 8.14$; range = 55–82) and a mean of 14.9 years of formal education ($SD = 3.42$; range = 5–18). Regarding PPA variants, most individuals were diagnosed with lvPPA (9/17), while the remaining cases were classified as having svPPA (4/17) or nfvPPA (4/17). Demographic characteristics and estimated disease severity are summarized in Table 1. Disease severity scores were retrospectively estimated based on the available cognitive and functional clinical information provided by each institution.

The control sample consisted of 60 healthy individuals (45 women and 15 men), with a mean age of 67.7 years ($SD = 7.72$; range = 51–84) and a mean of 14.3 years of formal education ($SD = 3.51$; range = 5–18). These 60 controls were organized into 13 control groups (CG), each composed of seven individuals. Controls could contribute to more than one CG, and in some cases the same CG was used for comparisons with more than one participant when demographic characteristics were comparable. For example, CG5 was used for comparisons with participants P5 and P6, as they shared similar demographic characteristics. To assess the absence of statistically significant differences in age and years of education between each participant and their corresponding CG, a modified *t* test for single-case comparisons was performed (Crawford & Howell, 1998; Crawford et al., 2010; see Table 2). With respect to gender, each participant was matched with control individuals of the same gender.

2.2. Materials

Participants diagnosed with PPA and CG individuals received the same assessment battery, which comprised four tasks evaluating word and nonword reading and writing. Tasks previously used in studies aimed at identifying patterns of acquired dyslexia and dysgraphia in Spanish-speaking populations were selected (Difalci et al., 2021; Matías-Guiu et al., 2017; Morello-García et al., 2021; Wilson & Martínez-Cuitiño, 2012). None of the participants were familiar with or had been previously assessed using the selected tasks, as these measures were not employed for the clinical diagnosis of PPA nor for the classification of its variants.

The assessment instruments included in the battery are described below.

2.2.1. Word and nonword reading from the *transpruebas* battery (W-NW-Read)

The *Transpruebas* battery (China & Ferreres, 2017) was designed to examine the lexical and sublexical processes

Table 1 – Demographic data of participants diagnosed with PPA.

ID	Age (years)	Education (years)	Education (level)	PPA variant	Estimated disease severity ^a
P1	79	12	Complete secondary education	Logopenic	NA
P2	55	16	Complete tertiary education	Logopenic	.5
P3	58	16	Complete tertiary education	Nonfluent	.5
P4	69	14	Incomplete university education	Nonfluent	.5
P5	69	17	Complete university education	Semantic	.5
P6	71	18	Complete university education	Logopenic	.5
P7	72	5	Incomplete primary education	Logopenic	1
P8	79	17	Complete university education	Semantic	NA
P9	60	18	Incomplete university education	Nonfluent	1
P10	82	18	Complete university education	Nonfluent	1
P11	70	12	Complete secondary education	Logopenic	.5
P12	55	13	Incomplete university education	Logopenic	.5
P13	74	12	Complete secondary education	Semantic	NA
P14	64	15	Incomplete tertiary education	Logopenic	NA
P15	67	18	Complete university education	Logopenic	.5
P16	67	18	Complete university education	Logopenic	.5
P17	62	15	Incomplete university education	Semantic	NA

PPA = primary progressive aphasia. NA = not available.

^a Estimated disease severity was retrospectively determined based on the available cognitive and functional clinical information provided by each institution. Severity levels ranged from 0 (no cognitive-functional involvement) to 3 (severe cognitive-functional involvement). A score of .5 indicated lower cognitive-functional involvement, whereas a score of 1 indicated greater cognitive-functional involvement.

Table 2 – Mean and standard deviation of age and years of education of the CG assigned to each participant (n = 7 per group), and results of the statistical comparisons (participant vs CG).

ID	PPA variant	Age (participant)	Mean age (CG)	Age comparison: participant vs CG		Education (participant)	Mean education (CG)	Education: participant vs CG	
			(SD)	t	p		(SD)	t	p
P1	Logopenic	79	76.9 (4.45)	.441	.674	12	12.6 (1.40)	-.401	.702
P2	Logopenic	55	57.7 (2.75)	-.918	.394	16	17.4 (1.13)	-1.159	.291
P3	Nonfluent	58	57.7 (2.75)	.102	.922	16	17.4 (1.13)	-1.159	.291
P4	Nonfluent	69	66.7 (2.63)	.818	.445	14	14.3 (1.25)	-.224	.830
P5	Semantic	69	70.1 (3.53)	-.291	.780	17	17.6 (.535)	-1.049	.335
P6	Logopenic	71	70.1 (3.53)	.238	.819	18	17.6 (.535)	.699	.511
P7	Logopenic	72	71.4 (4.20)	.134	.898	5	7.14 (1.35)	-1.483	.189
P8	Semantic	79	78.9 (3.53)	.026	.980	17	17.4 (.787)	-.475	.651
P9	Nonfluent	60	59 (4.04)	.232	.825	18	17.9 (.378)	.247	.813
P10	Nonfluent	82	78.9 (3.53)	.821	.443	18	17.4 (.787)	.713	.503
P11	Logopenic	70	66.1 (5.27)	.692	.515	12	13 (1.41)	-.663	.532
P12	Logopenic	55	58.7 (3.86)	-.897	.404	13	13.3 (1.38)	-.203	.846
P13	Semantic	74	72.6 (3.87)	.338	.747	12	12.1 (.900)	-.104	.921
P14	Logopenic	64	65 (3.87)	-.242	.817	15	15.4 (2.07)	-.181	.863
P15	Logopenic	67	67 (2.65)	.000	>.999	18	17.7 (.488)	.575	.586
P16	Logopenic	67	67 (2.65)	.000	>.999	18	17.7 (.488)	.575	.586
P17	Semantic	62	64.4 (4.16)	-.540	.609	15	15.4 (.535)	-.699	.511

PPA = primary progressive aphasia. CG = control group. SD = standard deviation.

involved in word retrieval. The W-NW-Read task comprises a total of 90 stimuli, including 45 words and 45 nonwords. The word stimuli consist of concrete, imageable nouns for which length, frequency, and syllabic complexity were controlled (Difalcis, 2023). The 45 nonwords were constructed by recombining the syllables of the word stimuli. Stimulus

length, stress pattern, and syllabic complexity were controlled, as well as syllable and phoneme frequency.

Participants were asked to read aloud each stimulus presented individually on a computer screen. Only reading accuracy was considered; therefore, task performance was coded dichotomously as correct or incorrect. In addition,

participants' responses were recorded in order to document the presence of semantic paralexias.

The complete list of stimuli for this task is available in the work published by [China and Ferreres \(2017\)](#).

2.2.2. Loanword reading (LW-Read)

The LW-Read task ([Wilson & Martínez-Cuitiño, 2012](#)) consists of a total of 40 loanwords in English and French that are frequently used in Argentina. The stimulus set includes 40 nouns for which familiarity, bigram frequency (token and type), orthographic neighborhood size, and word length (in letters) were controlled.

Participants were asked to read aloud each loanword presented individually on a computer screen. Only reading accuracy was considered.

The complete list of stimuli used in this task, as well as the familiarity indices for each item, are available in the work published by [Wilson and Martínez-Cuitiño \(2012\)](#).

2.2.3. Visual lexical decision with pseudohomophones (PSH-Decision)

The PSH-Decision task ([Difalcis, 2023](#); [Difalcis et al., 2018](#)) comprises a total of 159 stimuli: 53 words, 53 nonwords, and 53 pseudohomophones.

For task construction, two preliminary lists (A and B) of 55 nouns without stress marks were created. Lexical frequency, orthographic neighborhood size, word length (in letters), and bigram frequency (token and type) were controlled. Data used for matching these variables were obtained from the *BuscaPalabras* database ([Davis & Perea, 2005](#)). Half of the words in each list were of high frequency and the remaining half of low frequency. The 55 words from List A were included as part of the experimental word list. List B was used to generate the 55 nonwords and 55 pseudohomophones by substituting a single letter in each word. For example, from the List B word *cerebro* ("brain"), the nonword *verebro* and the pseudohomophone *serebro* were created. Finally, to construct the definitive stimulus set, six items (two from each stimulus type) that did not meet the minimum accuracy criterion established by the author were excluded ([Difalcis et al., 2018](#)), resulting in a final list of 53 words, 53 pseudohomophones, and 53 nonwords presented in a pseudo-randomized order.

Participants were instructed to respond "yes" if they believed the presented stimulus was an existing word in their language and "no" if they believed it was not. As in the previous tasks, stimuli were presented individually on a computer screen. Task performance was scored dichotomously as correct or incorrect.

The complete list of stimuli used in this task is available in the studies published by [Difalcis et al. \(2018\)](#) and [Difalcis \(2023\)](#).

2.2.4. Writing to dictation (Dictation)

The Dictation task comprised 60 stimuli: 40 words and 20 nonwords ([Morello-García et al., 2021](#)).

For the construction of the word list, regularity, lexical frequency, and word length were considered. With respect to regularity, 20 words were regular and fully consistent, meaning that they could be spelled correctly through phoneme-to-grapheme conversion rules (e.g., *materia*

"matter" and *alta* "high"), whereas the remaining 20 words were irregular, with varying degrees of inconsistency; that is, their correct spelling could not be derived from phonology alone and required knowledge of the word's specific orthographic representation (e.g., *pulsera* "bracelet" and *sapo* "toad"). Both regular and irregular words were balanced for lexical frequency and length. Lexical frequency data were obtained from the *BuscaPalabras* database ([Davis & Perea, 2005](#)).

Nonwords were created by recombining the syllables of 10 regular and 10 irregular words from the previously described word list. The selected base words were balanced for frequency and length. During nonword construction, syllable position within the base words was preserved.

Participants were asked to write on a sheet of paper the words dictated by the examiner. In the present study, regular and irregular words were dictated first, presented in a pseudo-randomized order, followed by the nonwords. One point was awarded for each correct response. In addition, participants' written responses were recorded for subsequent error analysis.

The list of stimuli included in this task is available in the study published by [Morello-García et al. \(2021\)](#).

2.3. Procedure

Individuals diagnosed with PPA were recruited at the participating healthcare institutions by their treating neurologists, who contacted the research team whenever they identified a case who met the inclusion and exclusion criteria. As part of routine clinical care, cognitive neurologists conducted a comprehensive clinical evaluation, including a detailed medical history, in which relevant information regarding exclusion criteria (e.g., history of acquired neurological conditions, neurodevelopmental disorders, and visual or auditory impairments) was routinely assessed. Subsequently, a brief interview was conducted by the research team to verify inclusion and exclusion criteria, and informed consent was obtained. Demographic information was also collected during this session. Given the age of the participants, potential neurodevelopmental disorders, such as dyslexia, were further assessed through retrospective questions regarding learning history, including difficulties during schooling, especially in reading and writing.

During the assessment, participants completed the tasks in the following order: PSH-Decision, W-NW-Read, LW-Read, and Dictation. The first three tasks were administered on a laptop computer. Individuals were seated approximately 45 cm from the screen. All stimuli were presented in lower-case letters, using Arial font, size 30. As participants responded to each stimulus, the examiner recorded the response and manually advanced to the presentation of the next stimulus. No maximum presentation time or response time limit was imposed; however, only the participant's first response was considered, and self-corrections were not allowed.

For the dictation task, participants were provided with a blank sheet of paper and a pen. They were instructed that they could use their preferred handwriting style and were free to organize the available space on the sheet as they wished, either writing the words vertically (one below the other) or horizontally.

Across all tasks, the examiner monitored participants' performance and recorded their responses using data sheets specifically designed for this purpose.

CG individuals were recruited through flyers distributed at the participating healthcare institutions and followed the same procedure as the participants.

The present study was reviewed and approved by the FLENI Research Ethics Committee (approval code: 4632) and was conducted in accordance with the Declaration of Helsinki and local ethical guidelines.

2.4. Data analysis

Data analyses were conducted using R software (version 4.2; R Core Team, 2022), as well as the Singlims_ES program (Crawford et al., 2010; Crawford & Garthwaite, 2002). For all analyses, statistical significance was set at $p \leq .050$.

To identify patterns of acquired dyslexia and dysgraphia in each participant, total accuracy scores were first calculated for each individual on the two oral reading tasks (W-NW-Read and LW-Read) and on the dictation task. Subsequently, each participants's performance was compared with that of their corresponding CG. For this purpose, the mean and standard deviation of each CG were calculated for the relevant tasks, and participant performance was compared against the CG using the Singlims_ES program (Crawford et al., 2010; Crawford & Garthwaite, 2002), which implements a modified t test for single-case comparisons (Crawford et al., 2010; Crawford & Howell, 1998).

The criterion for the presence of acquired dyslexia was defined as significantly poorer performance relative to the CG on either of the two oral reading tasks. Cases in which participants were unable to complete a task due to reading difficulties were also classified as exhibiting acquired dyslexia. In turn, the criterion for the presence of acquired dysgraphia was defined as significantly poorer performance relative to the CG on the dictation task.

To classify subtypes of acquired dyslexia in participants identified as presenting the condition, both within-subject comparisons and comparisons against the CG were conducted. For within-subject comparisons, associations between participants' performance and stimulus type were examined in the W-NW-Read task (words vs nonwords) and in the PSH-Decision task (pseudohomophones vs nonwords). These analyses were performed using a chi-square (χ^2) test whenever its assumptions were met; otherwise, Fisher's exact test was applied.

Comparisons against the CG were conducted to contrast participant performance with that of their reference CG on the W-NW-Read, LW-Read, and PSH-Decision tasks. For these analyses, a modified t test for single-case comparisons was used (Crawford et al., 2010; Crawford & Howell, 1998), implemented through the Singlims_ES software (Crawford et al., 2010; Crawford & Garthwaite, 2002).

The criteria proposed by Difalci (2023) for identifying each subtype of acquired dyslexia, originally developed for cases with focal aphasia, were adopted, with specific modifications required for their application in individuals diagnosed with PPA. These modifications included the addition of participant performance on the LW-Read task, which was not

considered in the original proposal, and the exclusion of findings from homophone comprehension tasks. This latter decision was based on the multiple difficulties exhibited by individuals with PPA that extend beyond impairments in single-word reading decoding, including deficits in single-word comprehension in svPPA and sentence comprehension difficulties in nvPPA and lvPPA. The criteria used for the classification of each type of acquired dyslexia in the present study are presented in Table 3.

Table 3 – Criteria for the classification of acquired dyslexia subtypes in primary progressive aphasia.

Surface acquired dyslexia

1. No significant differences in accuracy between words and nonwords within the same participant on the word and nonword reading task (W-NW-Read).
2. Impaired performance on loanword reading (LW-Read), defined as reduced accuracy compared with the corresponding control group.
3. In visual lexical decision with pseudohomophones (PSH-Decision):
 - a. Worse performance on pseudohomophones than on nonwords within the same participant, and
 - b. Lower accuracy on pseudohomophones compared with the control group.

Phonological acquired dyslexia

1. On the word and nonword reading task (W-NW-Read), the presence of significant differences in accuracy between stimulus types within the same participant, with poorer performance on nonwords.
2. Preserved performance on loanword reading (LW-Read), defined as no significant differences compared with the control group.
3. In visual lexical decision with pseudohomophones (PSH-Decision):
 - a. No significant differences in accuracy between pseudohomophones and nonwords within the same participant, and/or
 - b. No significant differences compared with the control group in accuracy on pseudohomophones.

Deep acquired dyslexia

1. The participant meets the criteria for phonological acquired dyslexia.
2. Presence of semantic paralexias on the word and nonword reading task (W-NW-Read).

Mixed acquired dyslexia

1. On the word and nonword reading task (W-NW-Read), the presence of significant differences in accuracy between stimulus types within the same participant, with poorer performance on nonwords.
2. Impaired performance on loanword reading (LW-Read), defined as reduced accuracy compared with the control group.
3. In visual lexical decision with pseudohomophones (PSH-Decision):
 - a. Worse performance on pseudohomophones than on nonwords within the same participant, and/or
 - b. Lower accuracy on pseudohomophones compared with the control group.

Unspecified acquired dyslexia

The participant presents acquired dyslexia, but the pattern of impairment does not correspond to any of the four categories defined above.

W-NW-Read = word and nonword reading. LW-Read = loanword reading. PSH-Decision = Visual lexical decision with pseudohomophones.

To classify subtypes of acquired dysgraphia in participants identified as presenting the condition, both within-subject comparisons and comparisons against the CG were conducted. For within-subject analyses, associations between participants' performance and stimulus type in the Dictation task (irregular words vs nonwords) were examined using a chi-square (χ^2) test whenever its assumptions were met; otherwise, Fisher's exact test was applied.

In addition, in cases in which an association between stimulus type and performance was identified, specifically, poorer performance on irregular words compared with nonwords, an ad hoc within-subject analysis comparing regular and irregular words was conducted to confirm the presence of a regularity effect.

Comparisons against the CG were performed to contrast participant performance with that of their reference CG on the dictation of irregular words and nonwords, in order to rule out potential effects of years of education. For these analyses, a modified t test for single-case comparisons was used (Crawford et al., 2010; Crawford & Howell, 1998), implemented using the Singlims_ES software (Crawford et al., 2010; Crawford & Garthwaite, 2002).

The criteria used to identify each subtype of acquired dysgraphia were developed based on the cognitive model proposed by Ellis and Young (2014) and are provided in Table 4.

In addition, in order to obtain more detailed information regarding participants' performance on the Dictation task, a descriptive analysis of the errors produced by participants was conducted in cases in which individuals exhibited surface or phonological acquired dysgraphia profiles. Errors were classified into two broad categories. Lexical errors were defined as those in which the final output resulted in a real word. In contrast, non-lexical errors were those in which the participant's final production was a nonword. The criteria used for error classification were those developed by Miranda (2019) and are provided in Supplementary Information, Appendix I.

3. Results

3.1. Reading impairments

The pattern of acquired dyslexia, operationalized as impaired performance on one or both oral reading tasks, was identified in 16 of the 17 participants in the sample (94%). Of these cases, eleven showed significant differences relative to their corresponding CG on both tasks, whereas five showed significant differences on only one of them (see Table 5).

According to the criteria proposed for the identification of acquired dyslexia subtypes (modified from Difalci, 2023; see Table 3), five participants were classified as presenting surface acquired dyslexia (P3, P7, P8, P14, P17), two participants as presenting phonological acquired dyslexia (P4, P6), and three participants as presenting mixed acquired dyslexia (P1, P9, P13). Six individuals could not be classified into any of the established subtypes (P2, P5, P10, P11, P12, P16). The only pattern that was not observed in the sample was deep acquired dyslexia, as participants with phonological profiles (P4 and P6) did not exhibit semantic paralexias.

Table 4 – Criteria for the classification of acquired dysgraphia subtypes in primary progressive aphasia.

Surface acquired dysgraphia
1. Impaired performance on writing to dictation of irregular words: - Worse performance on irregular words than on nonwords within the same participant, with poorer performance on irregular words than on regular words, and - Lower accuracy on irregular words compared with the control group.
2. Preserved performance on writing to dictation of nonwords, defined as no significant differences compared with the control group.
Phonological acquired dysgraphia
1. Impaired performance on writing to dictation of nonwords: - Worse performance on nonwords than on irregular words within the same participant, and - Lower accuracy on nonwords compared with the control group.
2. Preserved performance on writing to dictation of irregular words, defined as no significant differences compared with the control group.
Deep acquired dysgraphia
1. The participant meets the criteria for phonological acquired dysgraphia.
2. Presence of semantic paraphasias in writing to dictation.
Mixed acquired dysgraphia
1. No significant differences between stimulus types within the same participant, with a similar number of correct and incorrect responses for both irregular words and nonwords.
2. Lower accuracy than the control group on both irregular words and nonwords.
Unspecified acquired dysgraphia
The participant presents acquired dysgraphia, but the pattern of impairment does not correspond to any of the four categories defined above.

They correctly read all words presented in the W-NW-Read task.

Table 5 presents participants' performance on each reading task, together with comparisons against their respective CG and within-subject analyses.

With respect to PPA variants, among participants classified as svPPA, two presented surface acquired dyslexia (P8, P17), one mixed acquired dyslexia (P13), and one unspecified profile (P5). Among participants with nvPPA, one case of surface acquired dyslexia (P3), one phonological acquired dyslexia (P4), one mixed acquired dyslexia (P9), and one unspecified profile (P10) were identified. Finally, within the lvPPA group, two participants presented surface acquired dyslexia (P7, P14), one phonological acquired dyslexia (P6), one mixed acquired dyslexia (P1), and four unspecified profiles (P2, P11, P12, P16).

3.2. Writing impairments

With respect to acquired dysgraphia, defined by impaired performance on Dictation task, this pattern was observed in 12 of the 16 participants who completed the task (75%), all of whom showed significant differences compared with their respective CG on this measure (see Table 6).

According to the criteria proposed for the identification of acquired dysgraphia subtypes (see Table 4), one participant was classified as presenting surface acquired dysgraphia (P8), one participant as presenting phonological acquired

Table 5 – Performance of each participant and their corresponding control group ($n = 7$) on the W-NW-Read, LW-Read, and PSH-Decision tasks, results of participant–control statistical comparisons, and within-subject analyses by stimulus type.

ID	PPA variant	Task	Total correct (participant)	Mean correct (CG)	Participant vs CG		Effect size (Zcc)	Participant performance (vs CG)	Accuracy by stimulus type	Within-subject comparison	Stimulus-type effect	Acquired dyslexia subtype
				(SD)	t	p	Point estimate (95% CI)			p		
P1	Logopenic	W-NW-Read	75	89.7 (.756)	–18.189	<.001	–19.444 (–30.197; –8.800)	Impaired	43 W/32 NW	.005 ^a	Lexicality effect	Mixed
		LW-Read	20	37.4 (1.62)	–10.047	<.001	–10.741 (–16.708; –4.823)	Impaired	-	-	-	
		PSH-Decision	34	48.7 (5.35)	–2.570	.042	–2.748 (–4.416; –1.049)	Impaired	34 PSH/52 NW	<.001 ^a	Pseudohomophone effect	
P2	Logopenic	W-NW-Read	63	89.9 (.378)	–66.568	<.001	–71.164 (–110.442; –32.309)	Impaired	37 W/26 NW	.021 ^a	Lexicality effect	Unspecified
		LW-Read	7	39.3 (.756)	–39.965	<.001	–42.725 (–66.313; –19.388)	Impaired	-	-	-	
		PSH-Decision	52	51.9 (1.86)	.050	.962	.054 (–.690; .793)	Preserved	52 PSH/53 NW	>.999 ^b	No association	
P3	Nonfluent	W-NW-Read	85	89.9 (.378)	–12.126	<.001	–12.963 (–20.150; –5.841)	Impaired	44 W/41 NW	.360 ^b	No association	Surface
		LW-Read	36	39.3 (.756)	–4.083	.006	–4.365 (–6.872; –1.851)	Impaired	-	-	-	
		PSH-Decision	44	51.9 (1.86)	–3.973	.007	–4.247 (–6.692; –1.794)	Impaired	44 PSH/53 NW	.003 ^b	Pseudohomophone effect	
P4	Nonfluent	W-NW-Read	81	89.4 (.535)	–14.687	<.001	–15.701 (–24.393; –7.092)	Impaired	45 W/36 NW	.003 ^b	Lexicality effect	Phonological
		LW-Read	39	37.9 (1.46)	.705	.507	.753 (–.119; 1.580)	Preserved	-	-	-	
		PSH-Decision	47	50 (3.37)	–.833	.437	–.890 (–1.755; .024)	Preserved	47 PSH/53 NW	.027 ^b	Pseudohomophone effect	
P5	Semantic	W-NW-Read	86	89.9 (.378)	–9.651	<.001	–10.317 (–16.053; –4.628)	Impaired	44 W/42 NW	.616 ^b	No association	Unspecified
		LW-Read	36	38.4 (1.13)	–1.987	.094	–2.124 (–3.488; –.718)	Preserved	-	-	-	
		PSH-Decision	47	51.7 (1.11)	–3.961	.007	–4.234 (–6.672; –1.788)	Impaired	47 PSH/50 NW	.488 ^b	No association	
P6	Logopenic	W-NW-Read	84	89.9 (.378)	–14.600	<.001	–15.608 (–24.250; –7.050)	Impaired	45 W/39 NW	.026 ^b	Lexicality effect	Phonological
		LW-Read	38	38.4 (1.13)	–.331	.752	–.354 (–1.107; .426)	Preserved	-	-	-	
		PSH-Decision	51	51.7 (1.11)	–.590	.577	–.631 (–1.429; .209)	Preserved	51 PSH/53 NW	.495 ^b	No association	
P7	Logopenic	W-NW-Read	69	87.1 (2.04)	–8.300	<.001	–8.873 (–13.818; –3.963)	Impaired	37 W/32 NW	.319 ^a	No association	Surface
		LW-Read	17	33.4 (2.64)	–5.811	.001	–6.212 (–9.710; –2.727)	Impaired	-	-	-	
		PSH-Decision	20	37.3 (5.65)	–2.864	.029	–3.062 (–4.889; –1.209)	Impaired	20 PSH/49 NW	<.001 ^a	Pseudohomophone effect	

P8	Semantic	W-NW-Read	73	89.6 (.787)	–19.730	<.001	–21.093 (–32.753; –9.551)	Impaired	39 W/34 NW	.281 ^a	No association	Surface
		LW-Read	23	37.9 (1.35)	–10.324	<.001	–11.037 (–17.167; –4.959)	Impaired	-	-	-	
		PSH-Decision	40	51 (1.29)	–7.976	<.001	–8.527 (–13.284; –3.804)	Impaired	40 PSH/52 NW	.002 ^a	Pseudohomophone effect	
P9	Nonfluent	W-NW-Read	34	89.9 (.378)	–138.332	<.001	–147.884 (–229.496; –67.153)	Impaired	30 W/4 NW	<.001 ^a	Lexicality effect	Mixed
		LW-Read	17	39 (.816)	–25.220	<.001	–26.961 (–41.855; –12.221)	Impaired	-	-	-	
		PSH-Decision	41	51.9 (1.86)	–5.482	.002	–5.860 (–9.168; –2.562)	Impaired	41 PSH/45 NW	.456 ^a	No association	
P10	Nonfluent	W-NW-Read	50	89.6 (.787)	–47.068	<.001	–50.318 (–78.094; –22.838)	Impaired	26 W/24 NW	.832 ^a	No association	Unspecified
		LW-Read	22	37.9 (1.35)	–11.017	<.001	–11.778 (–18.314; –5.298)	Impaired	-	-	-	
		PSH-Decision	52	51 (1.29)	.725	.496	.775 (–.104; 1.608)	Preserved	52 PSH/52 NW	>.999 ^b	No association	
P11	Logopenic	W-NW-Read	84	88.7 (1.80)	–2.442	.050	–2.611 (–4.212; –.978)	Impaired	42 W/42 NW	>.999 ^b	No association	Unspecified
		LW-Read	12	37.1 (2.12)	–11.075	<.001	–11.840 (–18.410; –5.327)	Impaired	-	-	-	
		PSH-Decision	49	47.4 (4.47)	.335	.749	.358 (–.422; 1.111)	Preserved	49 PSH/53 NW	.118 ^b	No association	
P12	Logopenic	W-NW-Read	86	89.7 (.488)	–7.092	<.001	–7.582 (–11.823; –3.366)	Impaired	44 W/42 NW	.616 ^b	No association	Unspecified
		LW-Read	35	37.6 (3.05)	–.797	.456	–.852 (–1.706; .050)	Preserved	-	-	-	
		PSH-Decision	41	48.6 (2.07)	–3.434	.014	–3.671 (–5.813; –1.513)	Impaired	41 PSH/53 NW	<.001 ^a	Pseudohomophone effect	
P13	Semantic	W-NW-Read	83	89.9 (.378)	–17.075	<.001	–18.254 (–28.351; –8.257)	Impaired	W 45/38 NW	.012 ^b	Lexicality effect	Mixed
		LW-Read	7	37.6 (1.72)	–16.642	<.001	–17.791 (–27.633; –8.046)	Impaired	-	-	-	
		PSH-Decision	13	48.9 (6.04)	–5.560	.001	–5.944 (–9.296; –2.601)	Impaired	PSH 13/46 NW	<.001 ^a	Pseudohomophone effect	
P14	Logopenic	W-NW-Read	88	89 (1.15)	–.813	.447	–.870 (–1.728; .038)	Preserved	45 W/43 NW	.494 ^b	No association	Surface
		LW-Read	27	36 (2.77)	–3.039	.023	–3.249 (–5.172; –1.303)	Impaired	-	-	-	
		PSH-Decision	32	48.4 (3.10)	–4.949	.003	–5.290 (–8.291; –2.293)	Impaired	32 PSH/53 NW	<.001 ^a	Pseudohomophone effect	
P15	Logopenic	W-NW-Read	90	89.7 (.488)	.575	.586	.615 (–.221; 1.409)	Preserved	—	—	—	No evidence of
		LW-Read	37	38.7 (.951)	–1.672	.146	–1.788 (–2.997; –.531)	Preserved	-	-	-	acquired dyslexia
		PSH-Decision	—	—	—	—	—	—	—	—	—	
P16	Logopenic	W-NW-Read	83	89.7 (.488)	–12.843	<.001	–13.730 (–21.338; –6.192)	Impaired	43 W/40 NW	.434 ^b	No association	Unspecified
		LW-Read	30	38.7 (.951)	–8.557	<.001	–9.148 (–14.244; –4.090)	Impaired	-	-	-	
		PSH-Decision	45	52.4 (.787)	–8.796	<.001	–9.403 (–14.638; –4.207)	Impaired	45 PSH/51 NW	.097 ^a	No association	

(continued on next page)

Table 5 – (continued)

ID	PPA variant	Task	Total correct (participant)	Mean correct (CG)	Participant vs CG		Effect size (Zcc)	Participant performance (vs CG)	Accuracy by stimulus type	Within-subject comparison	Stimulus-type effect	Acquired dyslexia subtype
					(SD)	t	p					
P17	Semantic	W-NW-Read	81	89.7 (488)	–16.676	<.001	–17.828 (–27.690; –8.063)	Impaired	37 W/42 NW	.485 ^b	No association	Surface
		LW-Read	19	38.4 (1.40)	–12.962	<.001	–13.857 (–21.536; –6.250)	Impaired	–	–	–	–
		PSH-Decision	33	51.4 (2.30)	–7.483	<.001	–8.000 (–12.469; –3.560)	Impaired	33 PSH/53 NW	<.001 ^a	Pseudohomophone effect	–

PPA = primary progressive aphasia. CG = control group. SD = standard deviation. CI = confidence interval. W-NW-Read = word and nonword reading. LW-Read = loanword reading. PSH-Decision = Visual lexical decision with pseudohomophones. W = word. NW = nonword. PSH = pseudohomophone. ^a Statistical significance for the chi-square test. ^b Statistical significance for Fisher's exact test.

Note 1: For participant-CG comparisons in the PSH-Decision task, analyses were restricted to pseudohomophone items.

Note 2: Participant P15 did not meet the criteria to be classified as presenting acquired dyslexia, defined as impaired performance on one or both oral reading tasks.

dysgraphia (P10), and four participants as presenting mixed acquired dysgraphia (P2, P7, P11, P13). Six cases could not be classified into any of the established subtypes (P1, P3, P6, P12, P16, P17).

Deep acquired dysgraphia was not identified in the present sample, as the participant with phonological acquired dysgraphia (P10) did not exhibit semantic paraphasias.

Table 7 presents participants' performance on the Dictation task, together with comparisons against their respective CG by stimulus type and within-subject analyses.

For participant P8, and considering the observed findings, an ad hoc analysis was conducted to confirm the presence of a regularity effect. For regular words, the participant achieved an accuracy rate of 95% (19/20). When comparing performance on regular versus irregular words, a significant association between stimulus type and the participant's performance was found ($\chi^2 = 8.0251$, $df = 1$, $p = .005$). Accordingly, this participant exhibited a regularity effect, that is, greater difficulty in writing irregular words to dictation compared with regular words. Furthermore, a descriptive analysis of errors revealed that the majority of errors occurred in irregular words, for which 70% (7/10) were phonologically plausible non-lexical errors (e.g., *jubentud* for *juventud*, *bocablo* for *vocablo*) and 30% (3/10) were lexical stress-mark errors (e.g., *posicion* for *posición*, *buzon* for *buzón*).

In contrast, participant P10 performance was characterized predominantly by non-lexical errors. Of the errors produced on nonwords, 89% (8/9) were non-lexical spelling errors, mainly involving single-grapheme substitutions (e.g., *lapicen* for *lapigen*, *otina* for *otena*, *pulteso* for *pultedo*), whereas 11% (1/9) were lexicalization errors (e.g., *pegue* for *tegue*).

Finally, with respect to PPA variants, among participants classified as svPPA, one presented surface acquired dysgraphia (P8), one mixed acquired dysgraphia (P13), and one unspecified dysgraphia profile (P17). Among participants with nvPPA, one case of phonological acquired dysgraphia (P10) and one unspecified dysgraphia profile (P3) were identified. Finally, within the lvPPA group, three participants presented mixed acquired dysgraphia (P2, P7, P11), and four presented unspecified dysgraphia profiles (P1, P6, P12, P16).

Overall, the distribution of acquired dyslexia and dysgraphia profiles across PPA variants can be observed in Fig. 1.

4. Discussion

The aim of the present study is to examine reading and writing impairments in a sample of Spanish-speaking participants diagnosed with PPA. To this end, a single-case series approach was employed to identify participants presenting acquired dyslexia and dysgraphia and to classify them according to subtype.

Although reports of reading impairments in PPA, particularly surface acquired dyslexia, date back several decades and were initially described in individuals diagnosed with semantic dementia (Neary et al., 1998; Patterson & Hodges, 1992), the first report of acquired dyslexia in Spanish-speaking cases with PPA did not appear until 2017 (Matías-Guiu et al., 2017). To date, only four studies have examined reading impairments in Spanish-speaking individuals with

Table 6 – Performance of each participant and their corresponding control group ($n = 7$) on the Dictation tasks and results of participant–control statistical comparisons.

ID	PPA variant	Total correct (participant)	Mean correct (CG)	Participant vs CG		Effect size (Zcc)	Participant performance (vs CG)
			(SD)	t	p	Point estimate (95% CI)	
P1	Logopenic	46	57.4 (2.57)	–4.149	.006	–4.436 (–6.980; –1.885)	Impaired
P2	Logopenic	9	59.6 (.535)	–88.471	<.001	–94.579 (–146.777; –42.944)	Impaired
P3	Nonfluent	56	59.6 (.535)	–6.294	<.001	–6.729 (–10.507; –2.969)	Impaired
P4	Nonfluent	56	58.4 (1.51)	–1.487	.188	–1.589 (–2.713; –.417)	Preserved
P5	Semantic	58	58.7 (1.6)	–.409	.697	–.438 (–1.201; .358)	Preserved
P6	Logopenic	53	58.7 (1.6)	–3.332	.016	–3.563 (–5.648; –1.459)	Impaired
P7	Logopenic	29	52.7 (3.77)	–5.880	.001	–6.286 (–9.824; –2.762)	Impaired
P8	Semantic	49	58.7 (1.25)	–7.259	<.001	–7.760 (–12.098; –3.448)	Impaired
P9	Nonfluent	*	59.4 (1.13)	–	–	–	Not assessable
P10	Nonfluent	50	58.7 (1.25)	–6.510	<.001	–6.960 (–10.863; –3.077)	Impaired
P11	Logopenic	49	57.6 (2.88)	–2.793	.031	–2.986 (–4.775; –1.171)	Impaired
P12	Logopenic	52	57.9 (1.07)	–5.158	.002	–5.514 (–8.635; –2.399)	Impaired
P13	Semantic	32	56.4 (3.41)	–6.693	<.001	–7.155 (–11.165; –3.168)	Impaired
P14	Logopenic	50	56 (2.38)	–2.358	.056	–2.521 (–4.077; –.931)	Preserved
P15	Logopenic	58	59.4 (1.13)	–1.159	.291	–1.239 (–2.222; –.205)	Preserved
P16	Logopenic	53	59.4 (1.13)	–5.298	.002	–5.664 (–8.865; –2.469)	Impaired
P17	Semantic	46	58.3 (1.5)	–7.670	<.001	–8.200 (–12.778; –3.652)	Impaired

PPA = primary progressive aphasia. CG = control group. SD = standard deviation. CI = confidence interval. Dictation = writing to dictation.
 Note: * The participant was unable to complete the task due to her difficulties (ideomotor apraxia).

PPA (Aguillón-Solís et al., 2019; Matías-Guiu et al., 2017, 2019; Morello-García et al., 2021), two of which were single-case studies. In this context, the present study adds substantially to the existing literature by providing a systematic characterization of reading impairments in Spanish-speaking participants with PPA.

The situation is even more striking with respect to writing impairments, as there is a substantial gap in the Spanish-language literature. The presence of surface acquired dysgraphia in this population was reported several decades ago in studies of semantic dementia (for example, Neary et al., 1998) and, like surface acquired dyslexia, constitutes one of the supportive diagnostic features of PPA (Gorno-Tempini et al., 2011). However, to date, there has been only one comprehensive report of acquired dysgraphia in a Spanish-speaking individual diagnosed with PPA (Morello-García et al., 2021), along with three additional publications that did not provide sufficient detail regarding participants' performance (Donoso et al., 2008; Garzón Avellaneda et al., 2011; Reyes Nava, 2006). Consequently, the findings of the present study are highly relevant to the field of writing and its impairments.

Of the 17 participants evaluated, acquired dyslexia was identified in 16, and 10 of these participants could be classified into one of the dyslexia subtypes described in the literature, namely surface, phonological, or mixed dyslexia. Consistent with previous findings reported by Matías-Guiu et al. (2017), the presence of acquired dyslexia was observed across all PPA variants, indicating that reading impairments can be identified in each of the three most frequent PPA variants. These findings underscore the relevance of systematically examining reading impairments in Spanish-speaking individuals diagnosed with PPA.

With respect to acquired dysgraphia, this pattern was identified in 12 of the 16 participants who completed the

assessment. Six of these participants could be classified into established subtypes (surface, phonological, and mixed), whereas the remaining 50% could not be assigned to a specific subtype and were therefore classified as unspecified. As observed for reading impairments, the presence of acquired dysgraphia was also found across all PPA variants. This finding is consistent with previous reports by Faria et al. (2013) and Neophytou et al. (2019), indicating that acquired dysgraphia can be identified in each of the three PPA variants.

When examining performance by PPA subtype, all participants evaluated with svPPA presented acquired dyslexia. The predominant pattern was surface acquired dyslexia (2/4), which is expected given that this profile constitutes one of the supportive diagnostic features of this variant (Gorno-Tempini et al., 2011). According to the dual-route cascaded (DRC) model, this profile is interpreted as an impairment of the lexical reading route, with an overreliance on the non-lexical route (Coltheart, 2010; Coltheart et al., 2001; Ziegler et al., 2000). This pattern can be evidenced through tasks involving the reading of regular and irregular words and nonwords (Gorno-Tempini et al., 2011). However, in Spanish-speaking contexts, there is a risk of underdiagnosis due to the characteristics of the orthographic system, namely the absence of truly irregular words for reading, which necessitates the use of alternative tasks to identify this profile (Signorini et al., 2000). The tasks employed in the present study, such as W-NW-Read and PSH-Decision, have previously been shown to be effective in identifying surface acquired dyslexia in Spanish-speaking individuals with focal lesions (see, for example, Difalcis et al., 2021). Nevertheless, until now, there has been insufficient data regarding their functioning in Spanish-speaking participants diagnosed with PPA.

Our findings corroborate the proposal advanced by Ferreres et al. (2005), demonstrating that surface acquired

Table 7 – Performance of each participant and their corresponding control group ($n = 7$) on the Dictation task, including participant–control statistical comparisons and within-subject analyses for irregular words and nonwords.

ID	PPA variant	Stimuli	Total correct (participant)	Mean correct (CG)	Participant vs CG		Effect size (Zcc)	Participant performance (vs CG)	Within-subject comparison	Acquired dysgraphia subtype
				(SD)	t	p	Point estimate (95% CI)		p	
P1	Logopenic	IRRW	14	18.4 (1.81)	−2.274	.063	−2.431 (−3.943; −.883)	Preserved	>.999 ^a	Unspecified
		NW	15	19.3 (.756)	−5.320	.002	−5.688 (−8.902; −2.481)	Impaired		
P2	Logopenic	IRRW	4	20 (0)	—	—	—	—	.106 ^b	Mixed
		NW	0	19.6 (.535)	−34.269	<.001	−36.636 (−56.865; −16.621)	Impaired		
P3	Non-fluent	IRRW	18	20 (0)	—	—	—	—	>.999 ^b	Unspecified
		NW	19	19.6 (.535)	−1.049	.335	−1.121 (−2.062; −.130)	Preserved		
P6	Logopenic	IRRW	15	20 (0)	-	-	-	—	.182 ^b	Unspecified
		NW	19	18.7 (1.60)	.175	.867	.188 (−.568; .928)	Preserved		
P7	Logopenic	IRRW	6	16 (2.16)	−4.331	.005	−4.630 (−7.277; −1.978)	Impaired	.112 ^a	Mixed
		NW	12	17.9 (1.77)	−3.118	.021	−3.333 (−5.300; −1.346)	Impaired		
P8	Semantic	IRRW	10	19.7 (.488)	−18.593	<.001	−19.877 (−30.868; −8.997)	Impaired	.005 ^a	Surface
		NW	20	19 (1.41)	.663	.532	.709 (−.151; 1.525)	Preserved		
P10	Non-fluent	IRRW	20	19.7 (.488)	.575	.586	.615 (−.221; 1.409)	Preserved	.001 ^b	Phonological
		NW	11	19 (1.41)	−5.307	.002	−5.674 (−8.881; −2.474)	Impaired		
P11	Logopenic	IRRW	12	18.3 (1.98)	−2.976	.025	−3.182 (−5.070; −1.270)	Impaired	.068 ^a	Mixed
		NW	18	19.6 (.535)	−2.798	.031	−2.991 (−4.782; −1.173)	Impaired		
P12	Logopenic	IRRW	17	18.9 (1.21)	−1.469	.192	−1.570 (−2.686; −.406)	Preserved	>.999 ^b	Unspecified
		NW	17	19.1 (.690)	−2.847	.029	−3.043 (−4.861; −1.200)	Impaired		
P13	Semantic	IRRW	8	18.1 (1.68)	−5.624	.001	−6.012 (−9.401; −2.633)	Impaired	>.999 ^a	Mixed
		NW	8	18.9 (1.07)	−9.529	<.001	−10.187 (−15.851; −4.568)	Impaired		
P16	Logopenic	IRRW	17	20 (0)	-	-	-	—	>.999 ^b	Unspecified
		NW	17	19.4 (1.13)	−1.987	.094	−2.124 (−3.488; −.718)	Preserved		
P17	Semantic	IRRW	13	19.4 (.976)	−6.134	<.001	−6.557 (−10.242; −2.889)	Impaired	.479 ^a	Unspecified
		NW	16	18.9 (1.21)	−2.242	.066	−2.397 (−3.892; −.865)	Preserved		
PPA = primary progressive aphasia. CG = control group. SD = standard deviation. CI = confidence interval. IRRW = irregular word. NW = nonword. ^a Statistical significance for the chi-square test. ^b Statistical significance for Fisher's exact test.										

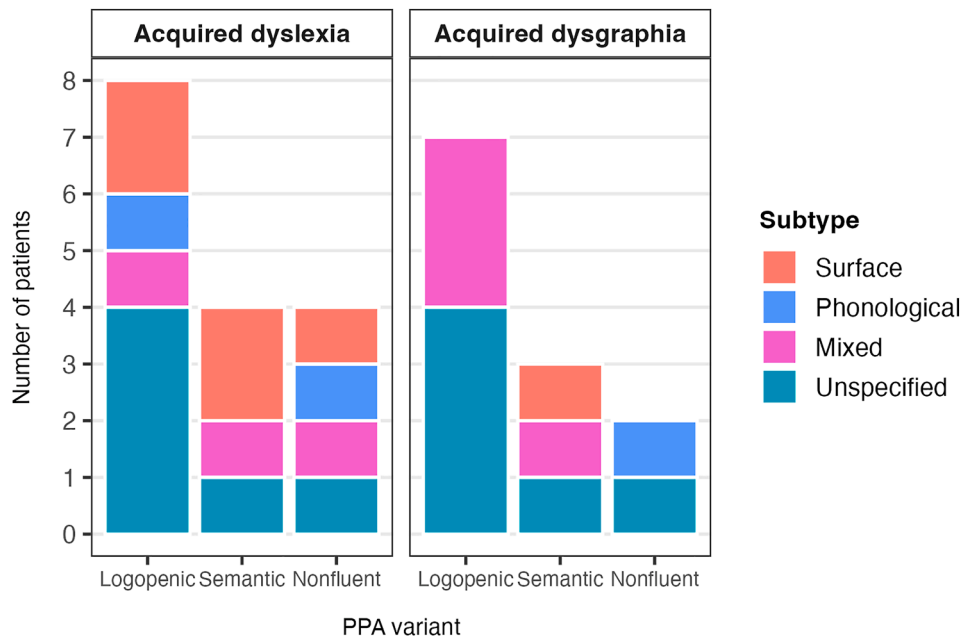


Fig. 1 – Distribution of acquired dyslexia and dysgraphia subtypes across primary progressive aphasia variants. The left panel shows acquired dyslexia subtypes among participants who met criteria for acquired dyslexia ($n = 16$), and the right panel shows acquired dysgraphia subtypes among participants who met criteria for acquired dysgraphia ($n = 12$). Bars represent the number of individuals within each PPA variant assigned to each subtype category.

dyslexia can be identified in Spanish when the linguistic characteristics of the language are appropriately considered during assessment. The present study represents one of the largest reports of surface dyslexia in Spanish-speaking populations to date, extending previous findings reported in both focal lesion and neurodegenerative contexts (Cuetos, 1999; Difalcis et al., 2021; Ferreres et al., 2005, 2012; Iribarren et al., 1996; Matías-Guiu et al., 2017; Morello-García et al., 2021). In addition, it is important to highlight that the tasks used in this study are easily applicable in clinical settings, as they do not require extensive resources. All three tasks can be administered either on a computer screen or in a paper-and-pencil format and are straightforward to implement.

With respect to writing performance in participants diagnosed with svPPA, only one individual did not present acquired dysgraphia. Notably, this group included the only case of surface acquired dysgraphia identified in the present sample. According to cognitive models of writing, this pattern is interpreted as an impairment of the lexical writing route, with a relative preservation of the non-lexical route (Ellis & Young, 2014). Our findings are consistent with previous reports describing the presence of surface acquired dysgraphia in this PPA variant (Faria et al., 2013; Henry et al., 2016; Mesulam et al., 2012; Neophytou et al., 2019; Sepelyak et al., 2011; Shim et al., 2012; Teichmann et al., 2019) and extend the single-case study reported by Morello-García et al. (2021) by further supporting the identification of this profile in Spanish-speaking participants with svPPA. These results demonstrate that surface acquired dysgraphia can be reliably identified in Spanish-speaking populations through the

administration of dictation tasks that include regular words, irregular words, and nonwords. Due to the presence of heterographic homophony, Spanish includes orthographically irregular words for writing, which enables the use of assessment tasks comparable to those traditionally applied in English-speaking populations.

With regard to participants diagnosed with lvPPA, the present study included nine individuals with this variant, making it the most frequent PPA subtype in the sample (9/17). Unlike svPPA, neither acquired dyslexia nor dysgraphia constitute supportive diagnostic criteria for lvPPA. Nevertheless, the presence of reading and writing impairments in this population has been reported in several studies (for example, Henry et al., 2016; Matías-Guiu et al., 2017; Ripamonti et al., 2017).

In our sample, only one participant with lvPPA did not present acquired dyslexia. Of particular interest were three cases, two classified as surface acquired dyslexia and one as phonological acquired dyslexia. The identification of the latter is particularly noteworthy, as it is consistent with previous reports describing phonological reading impairments in this variant (Brambati et al., 2009; Matías-Guiu et al., 2017; Petroi et al., 2014; Ripamonti et al., 2017; Rohrer, Rossor, & Warren, 2010). However, the presence of additional acquired dyslexia profiles in this group highlights the importance of conducting a comprehensive assessment of reading abilities, one that captures the functioning of both the non-lexical and lexical reading routes.

With respect to writing, performance in lvPPA was more heterogeneous, and among the established subtypes, only

mixed dysgraphia profiles could be identified. These findings are consistent with previous reports suggesting that individuals diagnosed with lvPPA may present impairments affecting both writing routes, lexical and non-lexical, resulting in the types of errors described here (Faria et al., 2013; Neophytou et al., 2019; Sepelyak et al., 2011). Likewise, the difficulty in identifying a clearly delineated dysgraphic pattern in the remaining participants is in line with the observations of Sepelyak et al. (2011), who highlighted the presence of acquired dysgraphia in individuals with lvPPA together with substantial challenges in defining a specific acquired dysgraphia subtype.

Overall, our results provide further evidence regarding lvPPA, a variant that was described later than the other two and is considered the most complex in terms of clinical characterization and classification (Henry et al., 2016; Neophytou et al., 2019).

Finally, four participants with nvfPPA were evaluated. As in lvPPA, reading and writing impairments do not constitute supportive diagnostic criteria for this variant; however, several studies have reported the presence of these difficulties in this population (for example, Matías-Guiu et al., 2017; Neophytou et al., 2019; Rohrer, Rossor, & Warren, 2010; Shim et al., 2012).

Multiple studies have suggested that phonological acquired dyslexia and dysgraphia constitute the most frequent profiles in this PPA variant (Matías-Guiu et al., 2017; Neophytou et al., 2019; Rohrer, Ridgway, et al., 2010; Rohrer, Ridgway, et al., 2010; Shim et al., 2012), however, other reports have identified alternative presentations (Faria et al., 2013; Rohrer, Rossor, & Warren, 2010; Tree et al., 2005). Our findings underscore the importance of systematically examining reading impairments in individuals diagnosed with nvfPPA, as despite the small sample size for this subtype in the present study, all participants exhibited acquired dyslexia, each with a distinct impairment profile. Notably, one case of surface acquired dyslexia and one case of phonological acquired dyslexia were identified.

With respect to writing performance, two of the participants with nvfPPA presented acquired dysgraphia. One of these participants represented the only case of phonological acquired dysgraphia identified in the sample. This pattern can be interpreted as an impairment of the non-lexical writing route, accompanied by an overreliance on the lexical route. This finding replicates previous reports by authors such as Neophytou et al. (2019) and Shim et al. (2012). In addition, our results highlight the importance of assessing writing abilities in individuals diagnosed with nvfPPA who do not present apraxic impairments.

The present study contributes to the understanding of PPA as well as to the characterization of reading and writing impairments in Spanish-speaking individuals with neurological conditions. As discussed above, one of the main contributions of this work is the identification of five cases of surface acquired dyslexia. At the same time, the study represents one of the largest case series addressing acquired dysgraphia in the Spanish-language literature. Within this context, the identification and detailed characterization of a case of phonological acquired dysgraphia is particularly noteworthy, as it adds to the Spanish-language literature only two previously

reported cases of this profile (Iribarren et al., 2001; Zilber et al., 2018).

Despite these contributions, the present study has several limitations that should be considered. First, with regard to the participant sample, individuals were selected using a non-probabilistic, purposive convenience sampling strategy, including only cases who were accessible through collaborating healthcare institutions. This sampling approach necessarily limits the representativeness of the studied sample. In addition, the assessment of exclusion criteria, such as neurodevelopmental history, relied on clinical report and did not include standardized instruments, which may have affected the accurate identification of such conditions. Likewise, disease severity was retrospectively estimated based on the clinical information available for each participant, as standardized severity measures were not systematically available across institutions. Future studies would benefit from incorporating standardized severity measures to better characterize the relationship between disease stage and patterns of reading and writing impairment in PPA. As noted above, PPA is considered a rare condition and, although the number of participants evaluated and the methodological design adopted were sufficient to address the study objectives, replication in larger samples is warranted, particularly including a greater number of individuals diagnosed with svPPA and nvfPPA. The limited number of participants per PPA variant restricts the generalizability of the present findings.

Second, future studies would benefit from the inclusion of a broader range of tasks specifically targeting writing abilities. Although previous evidence supports the use of dictation tasks involving regular words, irregular words, and nonwords for the identification of acquired dysgraphia and its subtypes, and this approach is recommended by the international consensus diagnostic criteria (Gorno-Tempini et al., 2011), the incorporation of additional writing measures would allow for a more detailed and fine-grained characterization of dysgraphic profiles. In addition, because the dictation task was administered at the end of the assessment protocol for all participants, potential fatigue or order effects cannot be ruled out. Counterbalancing task order across participants may help minimize these potential effects in subsequent research.

5. Conclusion

In summary, the present study highlights the relevance of examining both reading and writing impairments in Spanish-speaking participants with PPA. These impairments may be relatively frequent in this population and, in some cases, should be considered during the diagnostic process, as in the case of svPPA. Our results provide preliminary evidence that identifying patterns of acquired dyslexia and dysgraphia in Spanish-speaking contexts is feasible when the transparency of Spanish for reading and the presence of heterographic homophony in writing are taken into account.

Furthermore, the establishment of well-defined criteria that incorporate language-specific characteristics may contribute to the accurate identification of these disorders. A precise characterization of individuals' impairment profiles not only enhances our understanding of their cognitive

performance, but also facilitates the development of more appropriate and targeted intervention strategies.

Further studies involving larger samples will be important to replicate and extend the present findings.

CRedit authorship contribution statement

Morello-García Florentina: Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Difalcis Micaela:** Writing – original draft, Supervision, Methodology, Investigation, Data curation, Conceptualization. **Leiva Samanta Daniela:** Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Calandri Ismael:** Supervision, Data curation. **Hernández Micaela:** Formal analysis, Data curation. **Crivelli Lucía:** Supervision, Investigation. **Allegri Ricardo:** Supervision, Investigation, Conceptualization. **Ferreres Aldo:** Writing – original draft, Supervision, Methodology, Investigation, Data curation, Conceptualization.

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Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cortex.2026.06.003>.

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